BIOS103.0.2 Methods and Results Report (30%)

202627-BIOS103 - Introductory Practical Skills in Biosciences I

BIOS103 Methods and Results Assessment Rubric

Criteria	Ratings									Points
Experimental methods Assesses the description of experimental design and practical steps, including the information needed to understand and repeat the work.	**25.0 pts** **Exceptional** This aspect is at the highest standard. Marks in this category should be extremely rare.	**21.25 pts** **Outstanding** Provides an exceptionally clear, accurate and concise account of the experimental design and procedure. Identifies relevant samples, groups, controls, replication, materials and conditions, and presents the steps in a logical order with enough detail for the work to be repeated.	**18.75 pts** **Excellent** Provides a clear, accurate and well organised account of the experimental design and procedure. Includes nearly all relevant samples, groups, controls, replication, materials and conditions. The work could be repeated with very few uncertainties.	**16.25 pts** **Very Good** Describes the experimental design and all main procedural steps clearly. Includes most relevant samples, groups, controls, replication, materials and conditions, with only minor omissions or unnecessary detail. The work could largely be repeated.	**13.75 pts** **Good** Describes the main experimental steps, but some important details about the design, samples, groups, controls, replication, materials or conditions are missing or unclear. Repeating parts of the work would require some assumptions.	**11.25 pts** **Adequate** Gives a basic account of the procedure, but several details needed to understand the experimental design or repeat the work are missing or unclear. May include irrelevant detail or some errors.	**8.75 pts** **Marginal Fail** Provides an incomplete or confusing account of the experimental procedure. Important design features and procedural details are absent or inaccurate, making repetition difficult.	**5.0 pts** **Fail** Provides very little usable description of the experimental procedure. Major steps or design features are missing or seriously inaccurate; the work could not be repeated.	**0.0 pts** **Absent** No assessable description of the experimental methods.	**25.0**
Data processing and analysis methods Assesses the description of data organisation, processing, calculations, statistical analysis, assumptions and software.	**20.0 pts** **Exceptional** This aspect is at the highest standard. Marks in this category should be extremely rare.	**17.0 pts** **Outstanding** Gives a precise, complete and reproducible account of how data were organised, processed and analysed. Clearly identifies calculations and summary statistics, any exclusions or transformations, the statistical test or model and its relevant assumptions, and the software used. The choices are appropriate for the data.	**15.0 pts** **Excellent** Gives a clear and comprehensive account of data processing and analysis. Identifies the main calculations and summary statistics, relevant exclusions or transformations, the test or model and software. Assumptions and analytical choices are described accurately, with only minor omissions.	**13.0 pts** **Very Good** Describes the main data processing and analysis steps clearly, including the principal calculations, summary statistics, test or model, and software. Most relevant assumptions are addressed; a few details may be missing.	**11.0 pts** **Good** Describes the general analysis workflow, but omits some important details about calculations, data processing, exclusions, test or model, assumptions or software. Some choices may be insufficiently explained or contain minor errors.	**9.0 pts** **Adequate** Provides a basic outline of analysis, but several steps are unclear or missing. Important information about data processing, calculations, the statistical method, assumptions or software is incomplete or may contain errors.	**7.0 pts** **Marginal Fail** Analysis methods are incomplete or difficult to follow. Key processing steps or statistical methods are omitted or inappropriate, and errors undermine reproducibility.	**4.0 pts** **Fail** Provides very little usable description of data processing or analysis. Major methods are missing or seriously incorrect.	**0.0 pts** **Absent** No assessable description of data processing or analysis methods.	**20.0**

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Figures At least two distinct figures are required. Assess relevance, accuracy, clarity and compliance with the QS book figure guidance: labelled axes with units where relevant; no chart title; sequential numbering; descriptive caption below the figure, defining relevant symbols, variables and units.	**30.0 pts** **Exceptional** This aspect is at the highest standard. Marks in this category should be extremely rare.	**25.5 pts** **Outstanding** Includes at least two distinct, relevant and accurate figures. Each is clear and legible, has appropriately labelled axes including units where relevant, has no chart title, and has a sequential figure number and a descriptive, self-contained caption below it. Captions define symbols, variables and units needed to understand the figure.	**22.5 pts** **Excellent** Includes at least two relevant and accurate figures. Figures are clear, with appropriate axis labels and units, no chart titles, and correctly placed, numbered captions below. Captions are descriptive and define nearly all symbols, variables and units needed for interpretation.	**19.5 pts** **Very Good** Includes at least two useful figures, mostly accurate and legible. Axis labels and units are generally appropriate; figures have no chart titles and captions are below, numbered and mostly descriptive. A few minor formatting or caption details are missing.	**16.5 pts** **Good** Includes at least two figures, but one may be weakly chosen, unclear or contain errors. There are noticeable lapses in axis labels, units, title removal, caption position, numbering or caption detail.	**13.5 pts** **Adequate** Includes only one usable figure, or includes two or more figures with substantial problems in accuracy, legibility or required formatting. Captions or axis labels are incomplete and interpretation is limited.	**10.5 pts** **Marginal Fail** Includes one or more figures, but most are unclear, inaccurate or poorly formatted. Required axis information or captions are largely missing, making the figures difficult to interpret.	**6.0 pts** **Fail** Includes no usable figures, or the figures submitted are seriously inaccurate or uninterpretable.	**0.0 pts** **Absent** No figures submitted.	**30.0**
Summary table At least one summary table is required. Assess accuracy and relevance of summary statistics and compliance with the QS book table guidance: sequential numbering; descriptive caption above the table, defining relevant symbols, variables and units; sensible decimal places consistent with measurement precision.	**15.0 pts** **Exceptional** This aspect is at the highest standard. Marks in this category should be extremely rare.	**12.75 pts** **Outstanding** Includes at least one accurate, well organised summary table with appropriate descriptive statistics. It has a sequential table number and a clear, self-contained caption above it; relevant symbols, variables and units are defined. Values use a sensible number of decimal places consistent with measurement precision.	**11.25 pts** **Excellent** Includes at least one accurate summary table with appropriate statistics. Its numbered, descriptive caption is above the table and defines nearly all information needed to understand it. Decimal places are sensible, with only minor lapses in presentation.	**9.75 pts** **Very Good** Includes at least one mostly accurate and useful summary table. Caption is above and numbered; it is generally descriptive, and values are mostly presented to sensible precision. A few minor details may be missing.	**8.25 pts** **Good** Includes at least one summary table, but it has some errors or omissions in statistics, labels, units or precision. The caption may be vague, incorrectly numbered or placed, or may not define all symbols and variables.	**6.75 pts** **Adequate** Includes a table, but it is incomplete, difficult to interpret or contains several errors. Caption placement or numbering is incorrect or missing, and decimal precision may be unsuitable.	**5.25 pts** **Marginal Fail** The table is substantially incomplete or inaccurate, with major problems in its summary values, labels or formatting. The caption is absent or does not allow the table to be understood.	**3.0 pts** **Fail** No usable summary table is provided, or the table is seriously inaccurate and uninterpretable.	**0.0 pts** **Absent** No summary table submitted.	**15.0**

Criteria	Ratings									Points
Results narrative	**10.0 pts** **Exceptional**	**8.5 pts** **Outstanding**	**7.5 pts** **Excellent**	**6.5 pts** **Very Good**	**5.5 pts** **Good**	**4.5 pts** **Adequate**	**3.5 pts** **Marginal Fail**	**2.0 pts** **Fail**	**0.0 pts** **Absent**	**10.0**

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Assesses accurate, concise reporting of findings and clear references to figures and the summary table by number.	This aspect is at the highest standard. Marks in this category should be extremely rare.	Presents the main findings accurately, logically and concisely. Directs the reader to all relevant figures and the summary table by their correct numbers, and reports key numerical or statistical outcomes clearly without unsupported claims.	Presents findings clearly and accurately in a logical order. Refers to the relevant figures and table by number and reports key outcomes well, with only minor lapses in precision or concision.	Gives a mostly clear and accurate account of the main findings and refers to most figures and the table by number. May include some unnecessary detail or omit a minor result.	Reports several relevant findings, but the narrative may be uneven, contain some unnecessary detail, or omit references to some figures or the table. A few inaccuracies or unsupported statements may occur.	Provides a basic account of findings, but it is unclear or incomplete in places. Figure and table references may be missing or incorrect, and some statements may not match the displayed data.	Results narrative is superficial, difficult to follow or substantially incomplete. It rarely guides the reader to the figures and table, and contains errors in describing the findings.	Results narrative is largely absent or seriously inaccurate and does not communicate the findings.	No assessable results narrative.	

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GAI declaration	**0.0 pts** **2% deduction** GAI acknowledgement provided but contains insufficient relevant details (if GAI was used).	**0.0 pts** **5% deduction** No GAI declaration	**0.0 pts** **No issues** GAI declaration provided. Declaration has sufficient details (if GAI was used)	**0.0**
Academic Integrity	**0.0 pts** **Category 1: Poor academic practice** Mark for this assessment reduced to 40% Category 1: Poor academic practice Determined by marker	**0.0 pts** **Category 2, 3, or 4** Report to Academic Integrity Officer Report Concerns	**0.0 pts** **No academic integrity concerns** There are no academic integrity concerns with this assignment	**0.0**

Total points: 100.0